Personal Portfolio Website

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Goals and Background

Background:

After completing lots of job applications while seeking a post graduation role, I thought about the idea of having a website to directly show off my skills, and help put a name to a face when employers are looking at my resume. So I decided to create a personal portfolio website to show off all of my experience and skills

Goals:

Develop a high-performance Single Page Application (SPA) to serve as a centralized professional hub.

Demonstrate mastery of modern frontend frameworks (React) through a custom-built, not templated, solution.

Create a scalable platform that grows with my career, allowing for easy updates of new skills and projects.

Intellectual Merits

Component-Based Architecture:

- Designed using a modular React architecture where the User Interface is broken down into independent, reusable pieces (Components).
- This approach ensures code maintainability and allows for isolated testing of features like the navigation bar or project cards.

Dynamic State Management:

- Utilizes React's internal state to handle user interactions instantly without reloading the page, providing a "native app" feel.

Custom Implementation:

- Avoids "drag-and-drop" website builders in favor of hand-written code, demonstrating distinct engineering effort and understanding of the Document Object Model (DOM)

Broader Impacts

1

Talent Discovery

Streamlines the recruitment process by providing a transparent, accessible medium for evaluating technical talent

2

Open Access

The project adheres to web accessibility standards, ensuring information is available to all users regardless of device or ability.

3

Digital Sovereignty

Reduces reliance on closed platforms (like LinkedIn) by establishing a self-owned piece of the digital real estate.

4

Blueprint for Peers

Serves as an open-source reference for other students transitioning from academic theory to production-ready web development.

Design Specifications

1

System Overview

Architecture: Client-Side Rendered (CSR) Single Page Application.

Frontend: React (JavaScript/JSX).

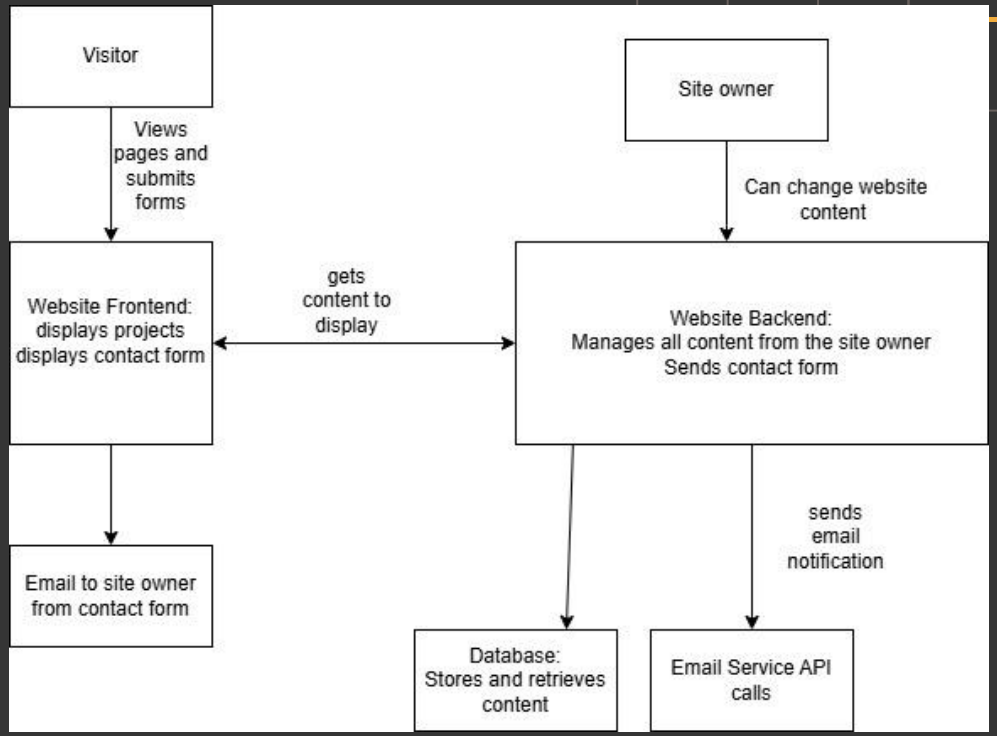
Routing: Virtual routing for seamless navigation between "Home," "Skills," and "Projects.

2

Implementation Details

Responsive Breakpoints: The layout automatically adjusts for Mobile (320px+), Tablet (768px+), and Desktop (1024px+).

Data Structure: Project data is decoupled from the UI, stored in structured JSON arrays to allow for easy updates without altering the source code



Technologies

1

Code Framework: React.js

Chosen for its Virtual DOM efficiency, which minimizes direct browser manipulation and improves performance and allows for declarative programming

2

Key Libraries

Marketing goals might include: reaching new audiences, generating new leads and sales, or strengthening existing customer relationships.

3

Development Tools

Version Control: Git & GitHub for source code management.

Package Manager: npm for dependency management.

Hosting: Netlify/Render for continuous deployment (CD).hips.

Milestones

Step 1

Design and Planning: Get a general idea of website layout and technologies needed

Step 2

Functionality Development: Develop backend of website and hosting service

Step 3

Stylizing: Finish all CSS and stylization of website. Fill in actual information into website, not just placeholders

Step 4

Testing: Test each and every functionality of website on different systems

Goal

Fully developed website, ready to be published

Results

1

Current Progress

I am currently on the stylization phase of development, fixing and updating all CSS to make sure the website looks nice and is smooth.

2

Completed Aspects

Currently have a working contact form, experience, about me, and skills section, working navbar ect.

3

Hosting of Website

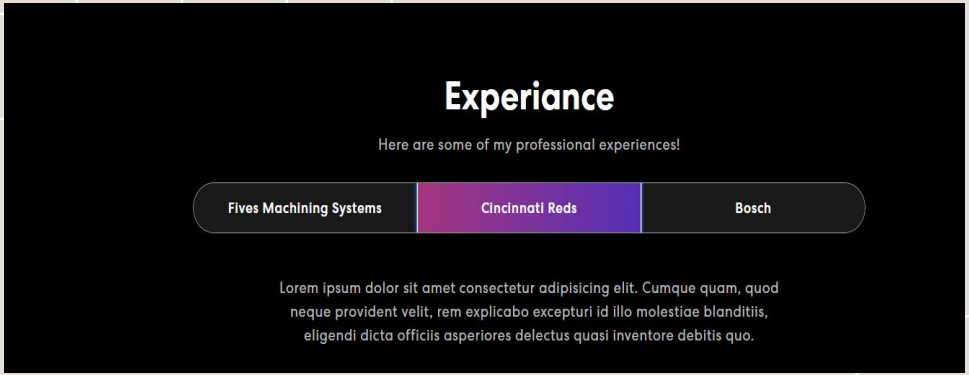
Have the backend for the website to be hosted finished, just need to purchase a domain and put the site up.

4

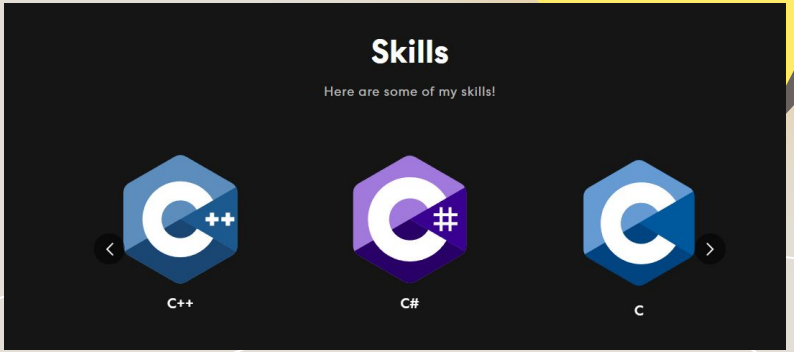
Testing

For the most part, I have been testing as I develop, so everything that I have created, works before final t

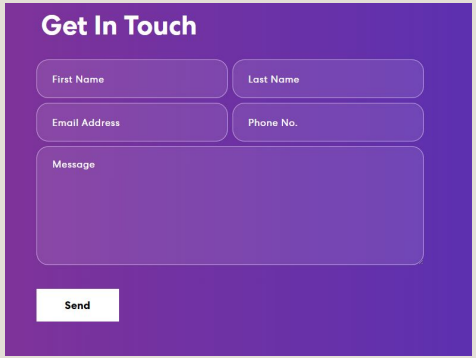
Artifacts



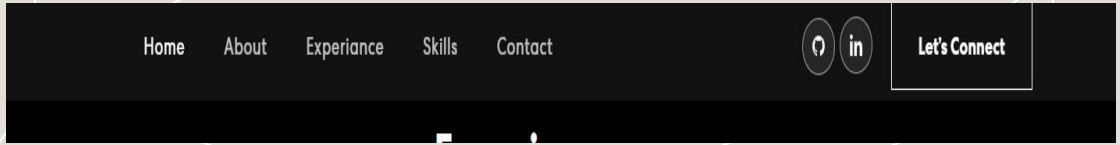
Experience selector



Skills carousel



Contact form



Navbar with working social media buttons

Challenges

Challenges

1. I have never done front end development work

   This entire process has been a big learning curve, not only leaning react, but CSS and front end development standards

2. Showcase professional experience without revealing proprietary code

   Adopted the "STAR" method (Situation, Task, Action, Result) to write abstract project summaries. This highlights the technologies and methodologies used, not sensitive data

3. Zero budget,creating a professional-grade hosting environment with no funding

   Leveraged the free tiers of modern PaaS providers (Netlify/Vercel) which offer enterprise-grade CDNs (Content Delivery Networks) at no cost for personal projects

Thank you